

MIMO System Simulation with BER Analysis

1. Aim

To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

2. Software Used

MATLAB

3. Code

```
clc;
clear;
close all;

% Number of bits
N = 10000;

% Generate random bits
data = randi([0 1],1,N);

% BPSK Modulation
tx = 2*data - 1;

% SNR Range
snr = 0:2:20;

BER = zeros(size(snr));

for k = 1:length(snr)

    % AWGN Noise
    rx = awgn(tx,snr(k),'measured');

    % BPSK Detection
    detected = rx > 0;
```

```
% BER Calculation
```

```

% Number of bits
N = 10000;

% Generate random bits
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% BPSK Modulation
tx = 2*data - 1;

% SNR Range
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BER = zeros(size(snr));

for k = 1:length(snr)

    % AWGN Noise
    rx = awgn(tx,snr(k),'measured');

    % BPSK Detection
    detected = rx > 0;

    % BER Calculation
    [~, BER(k)] = biterr(data,detected);

end

% BER Plot
figure;
semilogy(snr,BER,'-ob','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('MIMO System BER Analysis');

```

6. Output Graphs



7. Result

The MIMO system with BER analysis was successfully simulated using MATLAB. The BER performance was evaluated over different SNR values. The BER curve shows that as the Signal-to-Noise Ratio (SNR) increases, the Bit Error Rate (BER) decreases, demonstrating improved communication reliability and better performance of the MIMO wireless communication system.